NIHARIKA SAI

Joliet, IL |nihas19225@gmail.com | (443)- 931-2524 | LinkedIn |Github

(Open to Relocate)

SUMMARY

Results-driven **Business Analyst** with **3 years** of experience in **Data Visualization**, **ETL, Machine Learning,** and automation. Proficient in **Power BI, Tableau, SQL, Python,** and **AWS**, with a proven track record of delivering predictive analytics, real-time data pipelines, and automated reporting systems. Achieved up to **40%** efficiency gains while driving data-driven business decisions. Skilled in leveraging cloud platforms and advanced analytics to solve complex challenges and deliver actionable insights. **Actively seeking opportunities** to lead complex analytics projects and develop advanced business analysis strategies

AWARDS, RECOGNITION, CERTIFICATIONS

- Masters in SQL
- Automation anywhere-RPA
- AWS Cloud Practitioner
- Public AI/ML
- ISO 90001 Python
- Scrum Master Certification
- Received "Best Junior Employee" for the year 2021 for outstanding contribution to the data conversion team

PROFESSIONAL EXPERIENCE

ELEPHANT INSURANCE — **Baltimore, MD(Remote)**

Business Analyst — **May 2023 - Dec 2024**

The Client, a global insurance leader, faced challenges with outdated systems, inefficient data migration, and limited scalability in their data architecture. They needed robust solutions to support Guidewire systems, enable seamless cloud transitions, and drive business efficiency through scalable, flexible, and resilient data solutions.

- Designed and optimized insurance data models using Toad Data Modeler and **SQL** Server to support Guidewire PolicyCenter and ClaimsCenter, ensuring alignment with business needs
- Built robust **ETL** workflows in SSIS to transform and migrate policy, claims, and billing data from legacy systems to cloud environments like **AWS S3**
- Created real-time data pipelines using **Apache Kafka**, enabling continuous synchronization of policy and claims data across systems
- Developed predictive analytics models using **Python** and **AWS** SageMaker to forecast Guidewire PolicyCenter claims volume, detect fraudulent claims, and predict customer churn rates.
- Built AI-powered dashboards in **Tableau** and **Power BI**, allowing natural language queries to explore Guidewire policy and claims trends
- Virtualized Guidewire PolicyCenter data using Denodo, enabling real-time data insights without replication.
- Collaborated with **ETL** teams to design star and snowflake schemas, enhancing database query performance and scalability for Guidewire systems
- Developed **Python** scripts for data validation and reconciliation, ensuring accurate and error-free data migrations with minimal manual intervention
- Monitored and optimized pipeline performance using **AWS CloudWatch**, resolving bottlenecks to ensure seamless operations and efficient resource utilization
- Designed dynamic dashboards in **Tableau** and **Power BI**, delivering actionable insights into claims trends, policy performance, and financial metrics
- Utilized **Git, Jira**, and Confluence to ensure streamlined collaboration, version control, and comprehensive process documentation
- Recommended and integrated emerging technologies to future-proof the client's infrastructure, enhancing flexibility and resilience for evolving business needs

UMBC TRANSPORTATION SERVICES — **Baltimore, MD (Remote)**

Data Analyst — **Sep 2022 - Jan 2023**

The UMBC transit routes were designed years ago, but ridership patterns and community needs have significantly evolved. The lack of actionable insights from historical data and inadequate tools for real-time monitoring resulted in underutilized resources, inefficient route planning, and inconsistent service quality. These challenges led to delays, overcrowding, and declining customer satisfaction, highlighting the need for a data-driven approach to optimize transit operations.

- Conducted descriptive analytics to identify historical patterns and trends, such as ridership levels and peak usage times, improving service optimization by **20%.**

- Assisted in developing predictive analytics models for demand forecasting and traffic predictions using Python, enabling better resource allocation and route planning
- Recommended actionable insights through prescriptive analytics, optimizing scheduling efficiency, and improving route utilization by **15%**
- Built interactive dashboards and reports in **Tableau** and **Power BI,** enabling real-time monitoring and sharing insights with campus partners
- Performed geospatial analysis using GIS tools to map and analyze route efficiency and service coverage, enhancing decision-making for new route proposals
- Defined and tracked KPIs, including on-time performance and customer satisfaction, to measure and improve operational effectiveness
- Supported continuous improvement efforts by regularly reviewing operational strategies and incorporating data-driven insights to refine transportation services

PWC **Bangalore, India**
Role : Data Analyst / ETL (Also acted as developer and Scrum master) **Sep 2021 - Sep 2022**
Client: Western National Insurance (One of the largest insurance companies in the US)
Clients faced challenges with inefficient legacy systems, fragmented data processes, and a lack of scalable solutions, which hindered operational efficiency and decision-making. They needed streamlined applications, cloud-based scalability, optimized data workflows, and automated solutions to improve reliability, reduce manual errors, and accelerate business processes.

- Developed and optimized data-driven solutions to enhance client applications and streamline processes.
- Designed and implemented policy center and billing center application portals to meet client requirements, enhancing user experience and operational efficiency
- Conducted in-depth analysis of the client's legacy systems using **Toad Data Modeler** and **SQL Server** to identify data structure gaps critical for migration
- Extracted policy, claims, and billing data using SQL Server Integration Services (**SSIS**) and Python, ensuring data accuracy and compliance
- Designed and implemented **ETL** workflows in **SSIS** to **clean**, **transform**, and prepare data for migration into Guidewire InsuranceSuite
- Validated data accuracy and consistency through advanced **SQL queries** in **SSMS**, resolving discrepancies and maintaining data integrity
- Performed data mapping between legacy schemas and Guidewire data models to ensure seamless migration.
- Developed a new Rating feature for sales pitch presentations, enhancing client engagement
- Utilized SQL Server Data Tools (**SSDT**) for database management and development
- Worked extensively on data warehousing, **ETL** processes, and data modeling, consolidating data from multiple sources into centralized data warehouses
- Automated repetitive tasks using **Python** and **SQL**, reducing manual effort and improving accuracy
- Facilitated workshops and presentations to communicate project progress and deliverables effectively, fostering client confidence
- Managed projects using **Jira** and Confluence, coordinating sprints, resolving blockers, and tracking work updates
- Led and mentored junior team members, providing guidance to enhance their skills and achieve project goals

VALUELABS LLP **Hyderabad, India**
Data Intern **Jan 2020 – Aug 2021**
Client: Go-Ahead Ireland (Bus Transport Services)
The Client required insights into service-related data to answer critical questions about service performance, distribution, and trends. The existing process relied on monthly Excel sheet updates, and automating this process was crucial to improving report efficiency and accuracy.

- Created a sample report to generate insights based on the data provided by Go-Ahead Ireland.
- Established a connection to the data source using Power BI Gateway
- Configured the reporting tool to automatically pick up Excel sheets placed in a specific folder on the Go-Ahead server
- Scheduled regular data refreshes to ensure that updates or changes in the data were reflected in the Power BI reports
- Published the final report on Power BI Service, enabling stakeholders to access insights such as pending and completed services, service trends, distribution, and popular services

Client: Professional Risk Underwriting Pty Ltd (Insurance Claims Management Software)
The Client needed a Data Warehouse (DWH) and Business Intelligence (BI) solution to consolidate data from multiple platforms.

The goal was to present meaningful insights that would improve sales, revenue, and operational efficiency.
Tech Stack: SQL Server 2019, SSIS, SSDT, ETL, Power BI

- Built a DWH solution on SQL Server 2019 using the Snowflake model to consolidate data from multiple systems into a single source of truth
- Designed and implemented ETL services using SSIS to extract, transform, and load data from various insurance brokerage systems into the DWH
- Automated manual tasks through ETL jobs, increasing operational **efficiency by 20%**
- Enhanced data quality, achieving a **30% improvement** in the Data Quality Index by transitioning from manual Excel-based reporting to an automated DWH system
- Enabled a Power BI Dashboard on top of the DWH to provide actionable insights and significantly reduce the turnaround time for business-critical reports by 50%
- Delivered powerful BI reports showcasing impactful business insights, driving better decision-making and revenue growth

RESEARCH PUBLICATION

Enhancing Student Support with Deep Learning

- https://ieeexplore.ieee.org/document/9544747
- Developed a duplicate question detection system with deep learning using word2vec to provide needed services for student use.
- The new system aimed to be more efficient and effective than the current system, enabling better service delivery to users.
- Explored three deep learning approaches (DQ-CNN, DQ-RNN, and DQ-LSTM) based on CNN, RNN, and LSTM to solve the problem of duplicate question detection in Stack Overflow.

PROJECTS

Green Eco Analytica

- Engineered a custom preprocessing pipeline to clean and transform university datasets into SQuAD format Analyzed GHG emissions across 8 sectors using AWS Glue, Redshift, and QuickSight, identifying top emitting countries for policy insights.
- Streamlined data storage and processing with S3, enabling efficient queries and actionable visualizations.

DeepFake Audio Detection

- Built ML models on SageMaker to detect deepfake audio, ensuring digital media integrity.
- Automated workflows with Lambda and monitored model performance using CloudWatch for scalability.
- Developed a dynamic filtering system allowing users to search by make, model, price, year, and distance range, with real-time map updates

SKILLS

- **Databases**: Snowflake, Microsoft SQL Server, PostgreSQL, MongoDB(NoSQL)
- **Programming Languages**: Python (NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, Scipy), R (Shiny, dplyr, ggplot2), SQL, C, DAX, SQL, C, C++.
- **Data Visualization**: PowerBI, Tableau, Qlik Sense
- **Cloud**: AWS(EC2,ECS, S3, Lambda, API Gateway, Redshift, Glue, QuickSight,Sagemaker,CloudWatch), Azure(ADF, ADLS, Azure SQL, Azure Databricks), GCP(Dialogflow, Looker Studio)
- **Big Data**: Apache Spark, Apache AirRow, Hadoop, Databricks, Snowflake, Apache Kafka, Hive.
- **ETL**: SSIS, Azure Data Factory, AWS Glue
- **Tools**: Microsoft Excel, Docker, Streamlight, JIRA, Confluence
- CI/ CD: Git
- **Machine Learning**: Supervised ML (Linear and Logistic Regression, Decision Trees, RF, Xgboost), Unsupervised ML (K-means, PCA), NLP, Classification techniques (binary classification, multi-class classification), A/B testing
- **IDEs**: Visual Studio Code, PyCharm

EDUCATION

UNIVERSITY OF MARYLAND, BALTIMORE COUNTY **Maryland, US**
MS in Data Science (GPA: 3.6/4) **Sep 2022 - May 2024**

VIGNAN INSTITUTE OF TECHNOLOGY AND SCIENCE **Telangana, India**
B.Tech in Computer Eng. (GPA: 8.0/10.0) **Aug 2017 - May 2021**